Object for document set		
		For information
	X	For approval
		For Construction
		For installation
		For commissioning
		For maintenance

Status		
		Draft
		In review
		Approved
	X	As Built
		Released
		Withdrawn

Revision	Date	Description	Prepared	Approved
B	08-04-2015	As Commissioned	KAl	APr
A	15-12-2014	Revised after Client Comments	KAl	APr
-	11-11-2014	First Issue	KAl	APr

TP Aeolian Dynamics Wind Farm, 10.8 MW, at Agia Anna

Design, Manufacturing, Supply, Supervision of Erection and Commissioning of the MV switchgear and its associated control equipment, telecommunication equipment, load flow & short circuit study and earthing study of the 10.8MW Wind Farm in Ag. Anna, Larnaka, Cyprus.

Project: TP Aeolian Wind Farm 10.8MW at Agia Anna

Order No	Contract No	Substation	Format
P181114016		WIND FARM AGIA ANNA S/S	A4L
Prepared	**Responsible Dept.**	**Doc Kind**	**DCC**
KAl 11-11-2014		Signal list	&EMA
Approved	**Take Over Dept.**	**Title**	**Lang.**
APr 11-11-2014		System´s Signal list	en
	Derived from / Replaces	Ref. design. =E	

ABB SA	Document No.	Rev.	Sheet	No Sheets
ABB SA	3VGR111416F0002	B	1	7

Revisions				Client Doc No	Client	Project
			Date 11-11-2014		TP Aeolian Dynamics LTD	TP Aeolian Wind Farm 10.8MW at Agia Anna
B	8/4/2015	KAl	Prepared: KAl	ABB SA	Order No P181114016	Substation AGIA ANNA
A	15/12/2014	KAl	Approved: APr			Contract No
No	Date	Name				

Doc Kind	Signal list				
Title	System´s Signal list			Ref. design.	=E
Document No.	Rev.	Lang.	DCC	Page	Cont.
3VGR111416F0002	B	en	&EMA	1	2

No.	IED	BO/BI	LD	Logical Node	Data Ref		Object Text ABB	NCC1(IEC104) Address	IED HMI	Local HMI Event list/history	Local HMI Alarm list	LOCAL HMI(SCADA)	GATEWAY/NCC	RTU BO/BI	Status	Revision	Notes
1	AA1J1J01A200		LD0	GNRLCSWI1	PosOper.ctlVal		Circuit Breaker (QA1) Open/Close command	1601	x				x				
2	AA1J1J01A200		LD0	GNRLCSWI3	PosOper.ctlVal		BusBar Isolator (QB1) Open/Close command	2601	x				x				
3	AA1J1J01A200		LD0	GNRLCSWI1	PosstVal		Circuit Breaker (QA1) Open/Closed	16601	x				x				
4	AA1J1J01A200		LD0	GNRLCSWI3	PosstVal		BusBar Isolator (QB1) Open/Closed	17601	x				x				
5	AA1J1J01A200		LD0	GNRLCSWI4	PosstVal		Earth Switch (QC1) Closed	27601	x				x				
6	AA1J1J01A200					OR_3	Overcurrent Protection Operated(Inverse Time)	40601	x				x				
7	AA1J1J01A200		LD0	SPGGIO3	IndstVal		Earth Fault Protection Operated	42601					x				
8	AA1J1J01A200		LD0	SPGGIO4	IndstVal		Incomer J03 Protection Fault (IRF)	70603	x				x				
9	AA1J1J01A200		LD0	SPGGIO5	IndstVal		Circuit Breaker Fault	33601	x				x				
10	AA1J1J01A200		LD0	SPGGIO6	IndstVal		Trip Circuit Fault	30601	x				x				
11	AA1J1J01A200		LD0	SPGGIO10	IndstVal		General Bay Alarm	84601	x				x				
12	AA1J1J01A200		LD0	SPGGIO7	IndstVal		DC Supply Failure	86601					x				
14	AA1J1J01A200		LD0	SPGGIO1	IndstVal		CB Fail Prot, Operated	49601					x				
15	AA1J1J01A200		LD0	QCCBAY1	RemstVal		Supervisory Control Selected (NECC Authority)	58601	x				x				
16	AA1J1J01A200		LD0	PWRMMXU1	TotWmag.f		Active Power Flow (MWatt)	90601	x				x				
17	AA1J1J01A200		LD0	PWRMMXU1	TotVArmag.f		Reactive Power Flow (MVAr)	91601	x				x				
18	AA1J1J01A200		LD0	CPHMMXU1	A.phsBcVal.mag.f		Current (Amps)	92601	x				x				
19	AA1J1J01A200					OR_2	High set definite time earth fault protection operated	44601					x				

No.	IED	BO/BI	LD	Logical Node	Data Ref		Object Text ABB	NCC1(IEC104) Signal Type	NCC1(IEC104) Command Type	NCC1 (IEC104) Address	NCC1 (IEC101) Bit	IED HMI	Local HMI Event list/history	Local HMI Alarm list	LOCAL HMI(SCADA)	GATEWAY/NCC	RTU BO/BI	Status	Revision	Notes
1	AA1J1J02A200		LD0	GNRLCSWI1	PosOper.ctlVal		Circuit Breaker (QA1) Open/Close command		DC	1602		x				x				
2	AA1J1J02A200		LD0	GNRLCSWI3	PosOper.ctlVal		BusBar Isolator (QB1) Open/Close command		DC	2602		x				x				
3	AA1J1J02A200		LD0	GNRLCSWI1	PosstVal		Circuit Breaker (QA1) Open/Closed	DP		16602		x				x				
4	AA1J1J02A200		LD0	GNRLCSWI3	PosstVal		BusBar Isolator (QB1) Open/Closed	DP		17602		x				x				
5	AA1J1J02A200		LD0	GNRLCSWI4	PosstVal		Earth Switch (QC1) Closed	DP		27602		x				x				
6	AA1J1J02A200					OR_4	Overcurrent Protection Operated(Inverse Time)	SP		40602		x				x				
7	AA1J1J02A200		LD0	SPGGIO3	IndstVal		Earth Fault Protection Operated	SP		42602						x				
8	AA1J1J02A200		LD0	SPGGIO4	IndstVal		Feeder J01 Protection Fault (IRF)	SP		70601		x				x				
9	AA1J1J02A200		LD0	SPGGIO5	IndstVal		Circuit Breaker Fault	SP		33602		x				x				
10	AA1J1J02A200		LD0	SPGGIO6	IndstVal		Trip Circuit Fault	SP		30602		x				x				
11	AA1J1J02A200		LD0	SPGGIO10	IndstVal		General Bay Alarm	SP		84602		x				x				
12	AA1J1J02A200		LD0	SPGGIO7	IndstVal		DC Supply Failure	SP		86602						x				
14	AA1J1J02A200		LD0	SPGGIO1	IndstVal		CB Fail Prot, Operated	SP		49602						x				
15	AA1J1J02A200		LD0	QCCBAY1	RemstVal		Supervisory Control Selected (NECC Authority)	SP		58602		x				x				
16	AA1J1J02A200		LD0	PWRMMXU1	TotWmag.f		Active Power Flow (MWatt)	AM		90602		x				x				
17	AA1J1J02A200		LD0	PWRMMXU1	TotVArmag.f		Reactive Power Flow (MVAr)	AM		91602		x				x				
18	AA1J1J02A200		LD0	CPHMMXU1	A.phsBcVal.mag.f		Current (Amps)	AM		92602		x				x				
19	AA1J1J02A200					OR_5	High set definite time earth fault protection operated	SP		44602						x				

No.	IED	BO/BI	LD	Logical Node	Data Ref		Object Text ABB	NCC1(IEC104) Signal Type	NCC1(IEC104) Command Type	NCC1(IEC 104) Address	NCC1(IEC101) Bit	IED HMI	Local HMI Event list/history	Local HMI Alarm list	LOCAL HMI(SCADA)	GATEWAY/NCC	RTU BO/BI	Status	Revision	Notes
1	AA1J1J03A200		LD0	GNRLCSWI1	PosOper.ctlVal		Circuit Breaker (QA1) Open/Close-command		DC	1603		x				x				
2	AA1J1J03A200		LD0	GNRLCSWI3	PosOper.ctlVal		BusBar Isolator (QB1) Open/Close command		DC	2603		x				x				
3	AA1J1J03A200		LD0	GNRLCSWI1	PosstVal		Circuit Breaker (QA1) Open/Closed	DP		16603		x				x				
4	AA1J1J03A200		LD0	GNRLCSWI3	PosstVal		BusBar Isolator (QB1) Open/Closed	DP		17603		x				x				
5	AA1J1J03A200		LD0	GNRLCSWI4	PosstVal		Earthing Switch (QC1) Open/Closed	DP		27603		x				x				
6	AA1J1J03A200		LD0	VSGGIO2	DPCSOOper.ctlVal		A/R In/Out Command		DC	8603		x				x				
7	AA1J1J03A200		LD0	VSGGIO1	DPCSOOper.ctlVal		Lockout Relay Reset-command		SC	12603						x				
8	AA1J1J03A200		LD0	SPGGIO1	IndstVal		Overcurrent Protection Operated (Inv.Time)	SP		40603						x				
9	AA1J1J03A200		LD0	SPGGIO2	IndstVal		Earth Fault Protection Operated	SP		42603						x				
10	AA1J1J03A200		LD0	SPGGIO3	IndstVal		Feeder J02 Protection Fault (IRF)	SP		70602						x				
11	AA1J1J03A200		LD0	SPGGIO6	IndstVal		Circuit Breaker Fault	SP		33603						x				
12	AA1J1J03A200		LD0	SPGGIO5	IndstVal		Trip Circuit Fault	SP		30603						x				
13	AA1J1J03A200		LD0	SPGGIO10	IndstVal		General Bay Alarm	SP		84603						x				
14	AA1J1J03A200		LD0	SPGGIO7	IndstVal		DC Supply Failure	SP		86603						x				
15	AA1J1J03A200		LD0	QCCBAY1	RemstVal		Supervisory Control Selected (NECC Authority)	SP		58603		x				x				
16	AA1J1J03A200		LD0	SPGGIO15	IndstVal		CB Fail Protection Operated	SP		49603						x				
17	AA1J1J03A200		LD0	SPGGIO13	IndstVal		Auto Recloser Service Set Out	SP		65603						x				
18	AA1J1J03A200		LD0			OR_6	Under Frequency Protection Operated	SP		55603						x				
19	AA1J1J03A200		LD0	SPGGIO4	IndstVal		Sensitive Earth Fault Protection Operated	SP		43603						x				
20	AA1J1J03A200		LD0	DARREC1	PrgRecstVal		Auto Recloser Initiated	SP		78603						x				
21	AA1J1J03A200		LD0	DARREC1	LOstVal		Auto Recloser Lockout	SP		79603						x				
22	AA1J1J03A200		LD0	PWRMMXU1	TotWmag.f		Active Power Flow (MWatt)	AM		90603		x				x				
23	AA1J1J03A200		LD0	PWRMMXU1	TotVArmag.f		Reactive Power Flow (MVAr)	AM		91603		x				x				
24	AA1J1J03A200		LD0	CPHMMXU1	A.phsBcVal.mag.f		Current (Amps)	AM		92603		x				x				

No.	IED	BO/BI	LD	Logical Node	Data Ref		Object Text ABB	NCC1(IEC101) Response	NCC1(IEC104) Signal Type	NCC1(IEC104) Command Type	NCC1(IEC104) Address	NCC1(IEC101) Bit	IED HMI	Local HMI Event list/history	Local HMI Alarm list	LOCAL HMI(SCADA)	GATEWAY/NCC	RTU BO/BI	Status	Revision	Notes
1	AA1Y1Y01TH3					OR_1	Communication Equipment Fault		SP		78000						x				
2	AA1Y1Y01TH4	MIO:BI_5	LD0	SP16GGIO1	Ind5stVal		Fire Alarm Operated		SP		79000						x			B	
3	AA1Y1Y01TH4	MIO:BI_4	LD0	SP16GGIO1	Ind4stVal		Low Voltage AC Supply Failure		SP		82000						x			B	
4	AA1Y1Y01TH4	MIO:BI_1	LD0	SP16GGIO1	Ind1stVal		110V Battery Charger Fault		SP		87000						x			B	
5	AA1Y1Y01TH4	MIO:AI_1	LD0	MVGGIO1	AnInmag.f		MV room Temprature Measurement		AM		99600						x				
6	AA1J1J03A200		LD0	VPPMMXU1	PPV.phsABcVal.mag.f		22KV Busbar Voltage		AM		93600						x				
7	AA1Y1Y01TH4	MIO:BI_2	LD0	SP16GGIO1	Ind2stVal		110V Battery Charger Earth Fault		SP		85000						x				
8	AA1Y1Y01TH4	MIO:BI_3	LD0	SP16GGIO1	Ind3stVal		110V Battery Charger Over/Under voltage		SP		56000						x				
9	AA1Y1Y01TH3			SEV			S/S RTU OK		SP		84091						x				
10	AA1Y1Y01TH3						SCS Communicaton LAN Fault		SP		76000						x				

No.	Object Text ABB	NCC1(IEC104) Signal Type	NCC1(IEC104) Command Type	NCC1(IEC 104) Address	IED HMI	Local HMI Event list/history	Local HMI Alarm list	LOCAL HMI(SCADA)	GATEWAY/NCC	RTU BO/BI	Status	Revision	Notes
1	Activate/Deactivate Power Factor Control Mode		DC	12091					x				
2	Activate/Deactivate Reactive Power Factor Control Mode		DC	13091					x				
3	Overfrequency Curtailment Activated	SP		79091					x				
4	Reactive Power Control Mode activated	SP_OR_DC		80091					x				
5	Power Factor Control Mode activated	SP_OR_DC		81091					x				
6	Wind Farm Controller Failure	SP		77091					x				
8	Average (representative) Wind Speed (m/s)	AM		96091					x				
9	Average (representative) Wind Direction (degrees)	AM		97091					x				
10	Air Temperature (Degrees Celcius)	AM		99091					x				
11	No of Wind Turbines Available (RUNNING+PAUSED)	AM		16091					x				
12	No of Wind Turbines Unavailable (STOPPED-EMERGENCY+MANUAL)	AM		17091					x				
13	No of Wind Turbines in Unknown State(COMMS fault)	AM		18091					x				
14	Active Power Setpoint feedback (KW)	AM		90091					x				
15	Reactive Power Setpoint Feedback (KVar)	AM		100091					x				
16	Power Factor Setpoint Feedback 9Value:0,95 lead…0,95 lag)	AM		110091					x				
17	Active Power Setpoint (scaled value sent)	SPA		9091					x				
18	Power Factor Setpoint (scaled value sent)	SPA		11091					x				
19	Reactive Power Setpoint (scaled value sent)	SPA		10091					x				
20	Available Active Power (MW)	AM		92091					x				

No.	IED (FROM)	BO/BI	LD	Logical Node	Data Ref	GOOSE NAME	IED (TO)	GOOSE RECEIVE	GOOSE RECEIVE OUTPUT	Object Text ABB
1	AA1J1J01A200		LD0	SPGGIO14	IndstVal	GS01_03	AA1J1J03A200	GOOSEBINRCV:1	OUT_3	PHHPTOC BLOCK FROM DPHHPDOC START
2	AA1J1J01A200		LD0	SPGGIO15	IndstVal	GS01_03	AA1J1J03A200	GOOSEBINRCV:1	OUT_4	EFIPTOC BLOCK FROM DEFHPDEF START
3	AA1J1J01A200		LD0	SPGGIO1	IndstVal	gcb_A	AA1J1J03A200	GOOSEBINRCV:1	OUT_1	CB FAULT PROTECTION INTERTRIP
4	AA1J1J02A200		LD0	SPGGIO14	IndstVal	GS02_03	AA1J1J03A200	GOOSEBINRCV:1	OUT_8	PHHPTOC BLOCK FROM DPHHPDOC START
5	AA1J1J02A200		LD0	SPGGIO15	IndstVal	GS02_03	AA1J1J03A200	GOOSEBINRCV:1	OUT_9	EFIPTOC BLOCK FROM DEFHPDEF START
6	AA1J1J02A200		LD0	SPGGIO1	IndstVal	gcb_A	AA1J1J03A200	GOOSEBINRCV:1	OUT_2	CB FAULT PROTECTION INTERTRIP
7	AA1J1J03A200		LD0	SPGGIO16	IndstVal	GS03_0102	AA1J1J01A200	GOOSEBINRCV:1	OUT_1	PHHPTOC BLOCK FROM PHHPTOC START
8	AA1J1J03A200		LD0	SPGGIO16	IndstVal	GS03_0102	AA1J1J02A200	GOOSEBINRCV:1	OUT_1	PHHPTOC BLOCK FROM PHHPTOC START
9	AA1J1J03A200		LD0	SPGGIO17	IndstVal	GS03_0102	AA1J1J01A200	GOOSEBINRCV:1	OUT_2	EFIPTOC BLOCK FROM EFIPTOC START
10	AA1J1J03A200		LD0	SPGGIO17	IndstVal	GS03_0102	AA1J1J02A200	GOOSEBINRCV:1	OUT_2	EFIPTOC BLOCK FROM EFIPTOC START